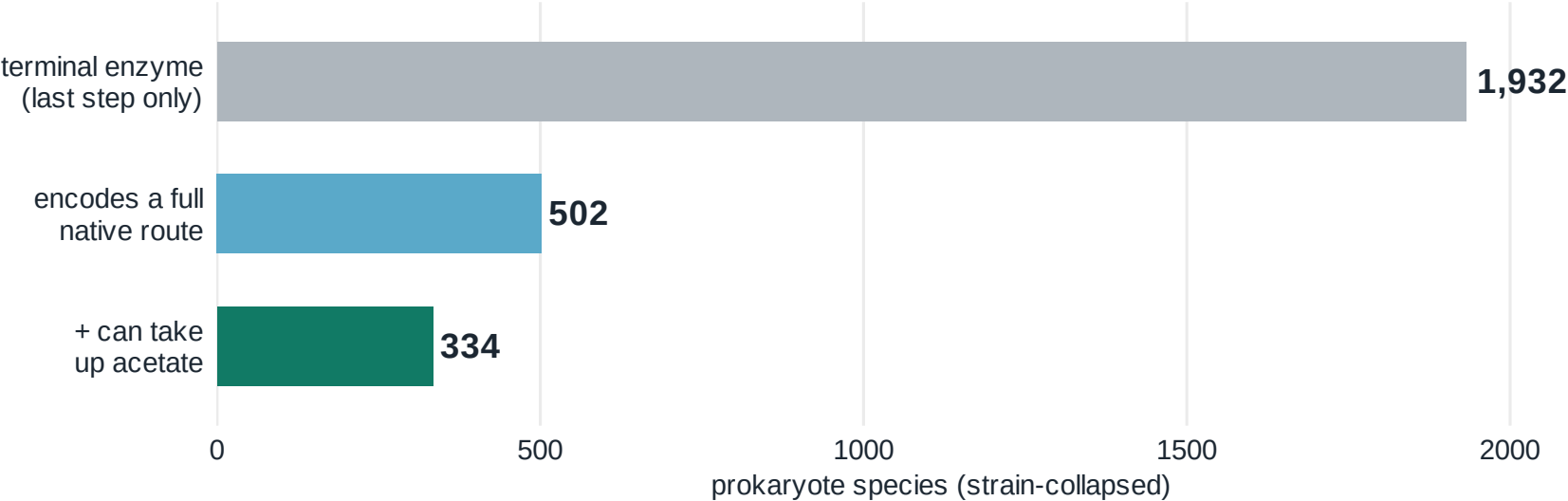


2,3-Butanediol from acetate: native capacity across KEGG prokaryote genomes

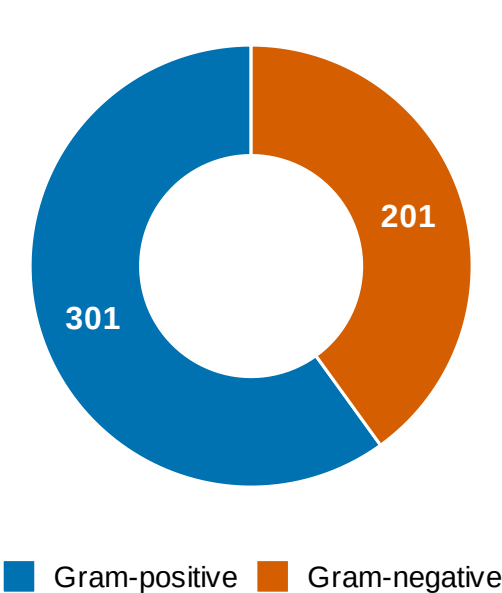
PanRoute retrosynthetic search · carbon-skeleton network · thermodynamically screened · direction-aware feedstock gating

a How many KEGG prokaryotes could make 2,3-butanediol from acetate

honest funnel — gene *presence*, not proof of production · 161 overflow-only species excluded



b Composition (n=502)



c Example route-carrying species

- *Bacillus subtilis* ✓ validated
- *Enterobacter cloacae* ✓ validated
- *Klebsiella pneumoniae* ✓ validated
- *Serratia marcescens* ✓ validated
- *Abyssicoccus albus*
- *Acetobacter orientalis*
- *Acetobacter persici*
- *Acetobacter senegalensis*
- *Acetobacter tropicalis*
- *Acinetobacter venetianus*
- *Actinobacillus delphinicola*
- *Actinobacteria bacterium*

Feedstock acetate → pyruvate · 16 routes (min 3 steps) · Validation P 0.80 / R 1.00

Genome *potential*, not production. A gene hit means the enzyme is encoded — not expressed, regulated, or carrying flux. KEGG genomes are culture-biased; Gram/taxonomy is heuristic. The terminal-enzyme count (1,932) overcounts because it ignores whether the cell can reach the precursor.